

SUMMARY

- My research interest lies in embodiment-agnostic task representation: extracting task structure from human demonstration video that transfers to agents with fundamentally different physical forms.
I am currently pursuing this through taskcraft, a self-directed research project using Minecraft as a low-cost, controllable pilot environment for cross-embodiment imitation via latent world models.

PROJECTS

- **taskcraft: Morphology-Agnostic Imitation from Human Video via Latent World Models**

Self-directed research project (Jul. 2026 – Present) investigating whether task representations extracted from human demonstration can transfer to agents with fundamentally different embodiments, using Minecraft as a low-cost pilot testbed.

- Positioned the project against existing cross-embodiment work (R3M, DreamGen, RT-X, NerveNet, MetaMorph): those methods generalize within visually/structurally similar embodiment sets (robot arm/gripper families, or randomly generated simulated morphologies that share a kinematic prior), while this project targets transfer across sets with no shared kinematic structure at all (human, robot, game avatar).
- Designed a falsifiable validation plan: linear-probe frozen human-video representations (R3M, VIP) against a Minecraft-native encoder (VPT) and a generic vision baseline (CLIP) on task-progress prediction before committing to full behavior cloning. R3M/VIP beating CLIP is evidence for the hypothesis; a null result bounds the claim rather than refuting it outright.
- Executing a pilot that transfers a frozen human-demonstration representation to a target agent whose control-to-dynamics mapping is qualitatively different within Minecraft (e.g. a camera axis that is irrelevant to movement while walking becomes the primary speed/altitude control while flying), benchmarked against a sparse-reward baseline.
- Identified an open validity threat and designed a check for it: a world-model latent trained only on human video may encode *how a human body caused a change* rather than *what changed in the world*, with no inherent training pressure to separate the two. Testing this by clustering human vs. Minecraft-avatar demonstration clips of the same task to see whether embeddings group by task or by embodiment.
- Repo: github.com/Akileox/taskcraft. Position paper and a 13-part technical blog series (VAE, World Models, Dreamer, MuZero, Genie, R3M, VIP) at akileox.github.io.

- **Dexterous Object Manipulation with PPO: Human-to-Robot Motion Retargeting**

Reinforcement Learning course final project (Isaac Lab, May 2026) training a simulated Shadow Hand to imitate human grasping motion extracted from real demonstration video.

- Given per-frame human hand keypoints and object pose reconstructed from multi-view video (HO-Cap), designed the observation/reward function so a PPO policy reproduces physically plausible manipulation instead of exploiting the tracking reward, including an explicit finger-group grasp reward and a Mixed RSI reset curriculum.
- Hand-engineered a human-to-robot correspondence (MANO keypoints to Shadow Hand) and an embodiment-specific reward for one fixed target embodiment: the same fully manual, non-reusable retargeting cost that motivates taskcraft’s embodiment-agnostic latent representation.
- Post: akileox.github.io/project/ppo-dexterous-manipulation

EMPLOYMENT

- **TeamDJ**

Product Planner & Full-stack Developer (Freelance)

Hybrid

May 2026 – Sep. 2026

- Planned and built, end-to-end, an LMS platform (dongdongmath.com) for a math academy, from feature design through deployment, as a paid freelance engagement in active production use.
- Designed an AI-assisted Q&A pipeline that reads a photographed math problem, returns an automatic first-pass answer with a generated explanatory diagram within 10 minutes, and lets the student accept it or escalate to a human teaching assistant.
- Designed and shipped automated learning-report generation from test/attendance data and automated review-video generation.

ACTIVITIES

- **Google Student Ambassador (#TeamGemini)**

2025

Designed and built "BioMBTI," a Gemini-powered (Imagen character art + real-time API classification) Arctic-wildlife personality-test campaign for environmental awareness on Instagram. Within one week of posting: 7,002 impressions, 53 likes, 77 comments.

- **Open Source Contribution Academy (OSSCA)**

2025

Merged a PR to PR Agent (Qodo, 12.8k-star open-source AI code review tool), modifying LangChainOpenAIHandler.

- PR: github.com/The-PR-Agent/pr-agent/pull/1828 (merged)

EDUCATION

- **Korea University**

B.S. in Computer Science (GPA: 4.45/4.5)

Seoul, South Korea

Mar. 2025 – Present

- All A+ across major coursework to date (Reinforcement Learning, Computer Architecture, and others)

- **Seoul National University**

Undergraduate Inter-University Credit Exchange, Digital Logic Circuits (A+)

Seoul, South Korea

Jun. – Aug. 2026

SKILLS

- **Programming** - Python, C/C++, Scala, Verilog (RISC-V)
- **ML / Simulation** - PyTorch, Isaac Sim
- **Tools** - Git, Docker, SQL

MILITARY SERVICE

- **Mandatory Military Service, Republic of Korea Air Force**

Sep. 2026 – Jun. 2028